Sahaj Bhadja

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Summary

Software engineer with 2+ years building maintainable automation, test frameworks, distributed workflows, and cloud services using Python, Java, PyTest, Mockito, Jenkins, GitHub Actions, Azure, AWS, and Docker.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Built fault-tolerant FastAPI workflows processing 5,000+ monthly jobs with explicit states, validation, reconciliation, failure isolation, and monitored releases, achieving 99%+ reliability across recurring production operations.
- Architected Azure SQL and Blob Storage services for 500+ users, monitoring query performance and tuning indexed access patterns to reduce dashboard latency from 1.2 seconds to 350 milliseconds.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented automated deployment and validation workflows for AWS-hosted Flask services handling 10,000+ daily requests, reducing feedback turnaround by 60% while maintaining stable performance during peak loads.
- Designed concurrent testable processing for 1,000+ uploaded files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, isolating failures and improving throughput across grading workflows consistently.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Automated nightly Jenkins pipelines for builds, VM installation, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, producing coverage summaries, CSV reports, and actionable failure logs.
- Troubleshot REST API and AWS Glue ETL workflows using S3 and SQL, improving response times by 30%, pipeline efficiency by 68%, and maintainability across distributed Agile teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting search, scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed web and mobile workflows under concurrent workloads.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Testing & Automation: PyTest, Mockito, unit testing, integration testing, regression testing, test automation, test design, validation, quality assurance, defect tracking, root-cause analysis, failure isolation; **Programming:** Python, Java, Bash, JavaScript, TypeScript, SQL, C++, C, C#, PowerShell; **Cloud & DevOps:** Azure, Azure SQL, AWS, Amazon S3, AWS Glue, AWS RDS, GCP, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, Git, monitoring; **Engineering:** REST APIs, FastAPI, Flask, microservices, distributed workflows, operational metrics, debugging, object-oriented design, Agile/Scrum, code reviews, performance optimization

Projects

Linguistix | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed a containerized data-processing API on AWS with automated CodePipeline releases, validation checks, and Docker builds, achieving 87% accuracy and production-grade reliability across repeatable integration test environments.

CPU-Scheduling Simulator Live | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% through deterministic simulation and systematic edge-case testing.

Chat2SQL Code | *Java, TypeScript, Docker, Maven, Node.js, Yarn*
- Built a cross-platform database workspace supporting 10+ systems, using Docker, Maven, Node.js, and Yarn to standardize maintainable development and deployment workflows across Windows, macOS, and Linux environments.